STAT_107_FINAL_PROJECT

Repository for STAT107 final project.

ABSTRACT

- **Intro: What is the central question that motivates your analysis? Who may this analysis benefit? Are you trying to train a model, test a hypothesis, classify objects apart? Why? This has to be written with proper grammar and syntax: sentences, paragraphs, and a narrative structure.**
  - For our project, we want to answer the question "Using one's genetic profile can we predict the possibility of tumors/breast cancer". We chose this as our topic of interest because with breast cancer being one of the most common forms of cancer and there being various existing databases containing genetic information, we were interested in further exploring the correlations and diving into a common medical practice with R. We'll be centering our research around specific genes that flag tumors and breast cancer more often such as TP53, BRCA1, BRCA2, CDKN2A, etc. Our goal is to extract the data, perform various statistical procedures such as linear or logistic regression with the genetic profile, and classify/return if the person of interest is at risk of any tumor or breast cancer, similar to an actual medical research center.
- **Executive Summary: provide, in a series of 3-5 bulletpoints, what are the core conclusions and findings of your analysis so far.**
  - So far, we have found that ???
- **Data: Where did you source the data, What variable/observations are included, Summaries of the variables, Cleaning**
  - We sourced the data from cbioportal.org's targeted sequencing of primary breast tumors. After cleaning it up, our data set is 449 observations of 9 different variables.
  - After cleaning up the data set, we decided on 9 variables to help answer our question. The variables and their significance are:
    - **ER_STATUS**: Whether or not the tumor cells have estrogen. These types of tumors respond best to hormonal therapy and generally have a better outlook for treatment
    - **GRADE**: The abnormality of the tumor under a microscope, a higher grade indicates faster growth and faster spread
    - **TMB_NONSYNONYMOUS**: Stands for tumor mutation burden and is a count of how many protein-changing mutations the tumor has
    - **TUMOR_SIZE**: The size of the tumor at the time of diagnosis in mm
    - **AGE_AT_DIAGNOSIS**: The age of the patient at the time of diagnosis
    - **LYMPH_NODES_EXAMINED_POSITIVE**: The number of nearby lymph nodes that contained cancer, a higher number indicates that the cancer has spread
    - **OS_BOOL**: Stands for overall survival status and is for if the patient has died. 0 means that the patient is living and 1 means that the patient has passed away
    - **RFS_BOOL**: Stands for recurrence-free survival status and is for if the cancer came back. 0 means that there was no recurrence and 1 means that there was a recurrence
    - **Recurrence**: Whether or not there was a recurrence of the cancer
- **Visualization: What do these visualizations suggest about the hypothesis, model or simulation you are aiming to perform? If you aim to generate further data visualizations, describe them here.**
  - Our scatter plot of 'Tumor size vs. positive lymph nodes' with the data points differentiated by if there was recurrence or not suggest that recurrence is typically indicated by a larger tumor size and more positive lymph nodes. Moreover, the box plot 'Positive lymph nodes by recurrence' suggest that no recurrence has a smaller cluster of positive lymph nodes

- **Analysis: Provide some preliminary plans of what analysis you aim to complete. Mention models, tests, procedures, and some ideas of what expected outcomes you hope to find.**
  - We plan to use ?? models to better answer our questions
- **Future work: Describe what is left to do and what is your plan**
  - We need to further analyze the results and flesh out the accuracy and cases of false positives, false negatives, etc.
- **Contributions**
  - Member 1 - Sri
    - Worked on the writing that explained our project details and created visuals (scatterplot and boxplot) that could help describe the relationship between several variables. The scatter plots could serve as supporting visuals for why there may or may not be relationships between the variables and will be further analyzed with models to better understand what the scatterplots could indicate. Defined the variables being used in the project and their significance to understanding our topic.
  - Member 2 did this []
  - Member 3 did this []

BE SURE TO INCLUDE THIS IN THE FINAL PROJECT AS WELL: • Data: What is the data available for performing the analysis? Where did you retrieve these data from? What are the variables and number of observations available? What steps where performed to clean the data? If the data is obtained via randomized simulations, clarify the data generating procedure. • Visualization: Provide preliminary visualizations in the form of histograms, density plots, scatter plots, box-plots, and numerical summaries of the data, depending on the type of analysis.